1. Free resolutions

① Over Noeth. local R, proj. $\Leftrightarrow$ free.

pf: See Thm 2.5 in MAT. (this holds $\forall$ local ring)

② Properties of minimal free resolutions.

- Min. res. is direct summand of any free res.
- Two min. res. are isomorphic.
- Min. res. has the shortest length of all res.

③ TFAE: for free res. $\cdots \to F_i \to F_{i-1} \to \cdots \to F_0 \to 0$.

i) $F_\cdot$ minimal

ii) $\partial F_i \subset \mathfrak{m} F_{i-1}$ (i.e the matrices of ∂ has entries in $\mathfrak{m}$). $\forall i \geq 1$.

iii) $\operatorname{rank} F_i = \dim_k \operatorname{Tor}_i^R(M,k)$ $\forall i \geq 0$.

iv) $\operatorname{rank} F_i = \dim_k \operatorname{Ext}_R^i(M,k)$ $\forall i \geq 0$.

} actually $\operatorname{Tor}_i^R(M,k) = F_i \otimes k = k^{\beta_i(M)}$.

pf: i) $\Leftrightarrow$ ii): for $\partial_i : F_i \to F_{i-1}$, consider:

$$\partial_i : F_i/\mathfrak{m}F_i \to \operatorname{Syz}_i(M)/\mathfrak{m}\operatorname{Syz}_i(M)$$

map of v.s.. $F_\cdot$ min. $\Leftrightarrow$ ∂_i sends basis to basis $\Leftrightarrow$ ∂_i isom.

$\Leftrightarrow \ker \partial_i \subset \mathfrak{m} F_i$.

ii) $\Leftrightarrow$ iii): Note that the complex: $\to = F_{i-1}/\mathfrak{m}F_{i-1}$

$$\cdots \to F_i \otimes k \xrightarrow{\partial_i \otimes id} F_{i+1} \otimes k \to \cdots$$

has $H^i(F_\cdot \otimes k) = k^{\beta_i(M)} \iff \partial_i \otimes id = 0 \iff \partial_i(F) \subset mF_{i-1}$

④ Definitions:

i) Betti no.: $\beta_i = \text{rank}(F_i)$

ii) $Syz_n(M) := \text{im}(\partial_n)$

iii) $\text{pdim}(M) :=$ length of any min. res. $= \sup\{i : Tor_i^R(M,k) \neq 0\}$.

iv) $\text{embdim}(R) := \mu(m)$.

⑤ Basic ineq. of pdim.

i) $\text{pdim}(M) \leq \text{pdim}(k)$.

pf: KEY is resolve both sides of $Tor_i^R(M,k)$:

- resolve k: then $H_n(M \otimes F_\cdot') = 0$ for $n = \text{pdim}(k)$.
- resolve n: then n is an upp. bound for $\{i : Tor_i^R(M,k) \neq 0\} \Rightarrow \text{pdim}(M) \leq n$.

ii) M summand of $N \Rightarrow \text{pdim}(M) \leq \text{pdim}(N)$.

pf: Then each term in min. res. is a summand.

2. Regularity.

By Krull's height,

$$\dim(R) = \text{height}(m) \leq \mu(m) = \text{embdim}(R).$$

R is regular if $\dim(R) = \text{embdim}(R)$. (othw. singular).

① The KEY to make every noeth. local ring regular:

Thm (Cohen): Every complete Noeth. local ring is a quotient of a regular local ring.

② Def: A minimal regular presentation of R is Q/I s.t.:

$\hat{R} \cong Q/I$, Q regular local, $I \subset m^2$.

Several facts building up to this:

i) Facts about sys. of param.

- ☆ Every Noeth. local admits a sys. of parameters.

Pf: See A.3. in B&H.

- $J = (x_1, \dots, x_n)$ for $n = \dim R$. Then:

$$J \text{ sys. of param.} \iff \sqrt{J} = m.$$

Pf: Note $\sqrt{J} = \bigcap_{P \supset J} P$, then by def. (min prime $\supset J$ is $m \iff \dim R/J = 0$)

- Every reg. seq. is part of sys. of param.
- ☆ R reg. $\iff$ m is generated by sys. of param.

Pf: "$\Rightarrow$" by def., since $\dim(R/m) = 0$; "$\Leftarrow$" then $\mu(m) = \dim R$.

ii) Application: facts about $\hat{R}$.

- $\dim R = \dim \hat{R}$

pf: Let J be a sys. of param.

Then $m^N \subset \sqrt{J} \Rightarrow m^N\hat{R} \subset J\hat{R} \Rightarrow m\hat{R} = \sqrt{J\hat{R}} = \bigcap_{P \supset J\hat{R}} P$

$\Rightarrow m\hat{R}$ is the min. prime. $\supset J\hat{R}$

$\xrightarrow{\text{Krull's}}$ $\dim \hat{R} = \text{height}(m\hat{R}) \leq \dim R$.

$\dim R \leq \dim \hat{R}$ by MAT ($\hat{R}$ is f.f.).

- R regular $\Leftrightarrow$ $\hat{R}$ regular.

pf: $\dim_k m/m^2 = \dim_k m\hat{R}/(m\hat{R})^2$. $(R/m \cong \hat{R}/m\hat{R})$.

iii) R regular local $\Rightarrow$ R int. dom.

pf: Induction on $\dim R$.

- $P_1, \cdots, P_n$ min. primes, m^2, by prime avoidance $\exists\, x \in m$ s.t. $x \notin$ all of them.
- Then x is part of min. gen. of m $\Rightarrow$ part of sys. of param. (see last • in i)).

$\Rightarrow \dim R/(x) < \dim R$, $R/(x)$ int. dom.

- Then (x) prime, $\exists\, P_i \subset (x) \Rightarrow$ for $z \in P_i$, $z = xy \in P_i \xrightarrow{x \notin P_i} y \in P_i$

$\Rightarrow P_i = xP_i \subset mP_i \xrightarrow{\text{NAK}} P_i = 0 \Rightarrow R$ int. dom.

iv) ✡✡ Thm: for $x \in R$ regular, $\begin{cases} \dim R/(x) = \dim R - 1, \\ \text{depth } R/(x) = \text{depth } R - 1. \end{cases}$

pf: $x \notin \text{Ass}(R) \Rightarrow$ avoids min. primes ; see 1.2.10 in B & H.

v) ✡✡ R regular, $f \in m$, then $R/(f)$ regular $\Leftrightarrow$ $f \in m \setminus m^2$.

pf: Since R domain, $f \in m$ is regular $\Rightarrow \dim R/(f) = \dim R - 1$.

Also $(m/(f))^2 = (m^2+(f))/(f)$, and $m/m^2 = m/m^2+(f)$.

Thus R regular $\Leftrightarrow \dim_k m/m^2+(f) = \dim_k m/m^2 - 1$

$\Leftrightarrow (f) \subset m^2$

since $f \notin m^2$ imposes a nontrivial relation on v.s.

Therefore, for $\hat{R} = Q/I$, we can write $Q/(f)/I/(f)$ for $f \in m \setminus m^2$

if $I \not\subset m^2$ & $m \neq 0$, then $Q/(f)$ still regular and $\dim Q/(f) < \dim Q$.

Thus the process will terminate until $I \subset m^2$.

③ $\text{embdim}(R) \overset{\downarrow}{=} \text{embdim}(\hat{R}) \overset{\downarrow}{=} \text{embdim}(Q)$.

pf: ②.ii) ②.v).

3. pdim to detect singularity.

① Thm (Auslander-Buchsbaum): TFAE for (R, m):

i) R regular

ii) k has finite proj. dim.

iii) Any f.g. R-mod. has finite proj. dim.

e.g. $k[[x]]$ & $k[[x]]/(x^3)$.

② Then we can solve the Localisation Problem for Regular Rings:

Thm: R is regular $\Rightarrow \forall$ prime P, R_P regular.

pf:

R regular $\Rightarrow R/P$ has fin. proj. dim.,

$$0 \to F_n \to F_{n-1} \to \cdots \to F_0 \to R/P \to 0 \quad \text{exact}$$

$\xRightarrow{-\otimes R_P}$ $0 \to F_n' \to F_{n-1}' \to \cdots \to F_0' \to R_P/PR_P \to 0$ exact

Since R_P flat.

Also $F_i' = \oplus R_P$ so is free $\Rightarrow$ R_P/PR_P has fin. proj. dim.

$\Rightarrow R_P$ is regular.

4. Depth

① Depth is well-def. for Noeth. local.

- $\operatorname{depth}(M) = 0 \iff m \in \operatorname{Ass}(M)$.

Pf: ☒ The reason for "local" R is that $M/xM \neq 0 \iff x \in m$. ($\overset{\text{by}}{\Rightarrow}$ NAK).

So the remaining condition is nonzerodiv. in m.

- So either $\operatorname{depth}(M) = 0$, or $m \notin P_i \in \operatorname{Ass}(M) \Rightarrow m \notin \bigcup_i P_i \Rightarrow$ choose x.

M f.g. $\Rightarrow$ eventually stop.

depth(M) := max. length of reg. seq. in m.

② Facts.

i) Thm (A-B formula):

$$\operatorname{pdim}(M) + \operatorname{depth}(M) = \operatorname{depth}(R).$$

pf: See 1.3.3 in B&H. (Note that $\operatorname{pdim}_{R/(x)}(M/(x))$ also decreases by 1).

ii) Thm: $\operatorname{depth}(M) = \min\{i : \operatorname{Ext}^i_R(M,k) \neq 0\}$.

pf: See 1.2.5 & things before in B&H.

iii) $\operatorname{depth}(R) \leq \dim(R)$

pf: reg. elt. decreases dim by 1.

5. Cohen-Macaulay.

R is C-M if $\operatorname{depth}(R) = \dim R$.

① Regular local ring is C-M.

Thm: R is regular $\Leftrightarrow$ m is generated by reg. seq.

pf: "$\Leftarrow$": $\operatorname{depth}(R) \leq \dim R \leq \operatorname{embdim}(R) \Rightarrow \dim R = \mu(m)$.

"$\Rightarrow$": Again use $R/(f)$ reg. for $f \in m \setminus m^2$.

Induction on $\mu(m)$.

If $m = (x_1, \cdots, x_n)$, then $R/(x_1)$ is still regular $\Rightarrow m = (\bar{x}_2, \cdots, \bar{x}_n)$ is generated by reg. seq. $\Rightarrow \bar{x}_2, \cdots, \bar{x}_n$ reg. on $R/(x_1) \Rightarrow (x_1, \cdots, x_n)$ reg. on R (independent of perm., or chose x_n; note $x_i \notin m^2$ as a basis elt.).

② Properties.

i) Thm:

R is C-M. Then:

$$\dim(R/I) = \dim R - \text{height } I.$$

Note that $\text{height } I = \inf\{\text{height}(\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec} R, I \subset \mathfrak{p}\}$.

(☆ minimum height for min. primes $\supset I$)

ii) ☆ Cor: R C-M. TFAE:

- I is gen. by reg. seq.
- Min. generating set of I is reg. seq.
- ☆ $M(I) = \text{height}(I)$ (Note always have $M(I) \geq \text{height}(I)$.